Solvency Field Theory V13.1-V3

What an Electron Withholds

Narrative companion to the V13.1-V3 Numerical G Companion

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> *The number the world calls G is not a property of gravity. It is the size of our kilogram compared to the universe's own mass scale. What SFT can now say is where the electron sits on that scale — and why it sits slightly lower than the law that allows it would suggest.*

Publication boundary. This companion narrates a result; it derives nothing new. Under one explicitly labelled postulate, SFT predicts the electron's dimensionless gravitational share and therefore numerical G, landing 0.976 parts per million above the current CODATA central value with zero continuous free parameters. It does not derive the muon, the tau, or any other particle. It does not prove the postulate. And it is not a one-part-per-million experimental confirmation, because measured G is itself uncertain at twenty-two parts per million. Every number below is taken from the frozen derivation canon.

1 You cannot weigh an electron

Every measurement of gravity ever made has been a measurement of two things against each other. Cavendish weighed lead against lead. A modern torsion balance weighs one sphere against another. There is no experiment that measures the gravity of a single object — and if gravity is relational, if what we call weight is the interaction of two warpings rather than a property either one carries by itself, then that is not a technological limit. There is nothing there to measure.

For an electron the limit is absolute. Between two electrons at the closest separation physics permits, the gravitational force is about 2.4×10^{-43} times the electric force between the same two particles at the same instant, in the same place. Moving them closer does not help: both forces scale identically, so the ratio never moves. The electron's gravitational mass has never been measured. It never will be.

So if a theory is going to say anything about an electron and gravity, it cannot check itself against a direct measurement. It has to derive the number, and then be tested somewhere else entirely.

2 What G actually is

Strip away the units and G stops looking like a force constant. In natural units it is simply the inverse square of the Planck mass. The familiar numeral 6.674×10^{-11} is not a statement about how strong gravity is; it is a statement about how large a kilogram is compared to the universe's own mass scale. Switch to grams and G changes. Gravity does not.

And that scale is not microscopic. The Planck mass is about 21.8 micrograms — a grain of pollen, something visible under a cheap microscope. It is the mass at which an object's gravity and its quantum behaviour become comparable, and it is the only mass scale in physics at which gravity can be isolated from everything else.

The electron sits a factor of 4.185×10^{-23} below it. That ratio is the real physical content of G. It is dimensionless, identical in every laboratory in the universe, and immune to any redefinition of the kilogram. Deriving it and deriving G are the same act; everything after it is arithmetic.

3 Why the equations could not produce it

For a long time the theory could not produce that number, and the reason turned out to be structural rather than a gap in the work.

Every law in the particle sector is a balance law: something restores, something resists, and the two meet. A balance does not care how large you draw it. A bridge design works whether the bridge is a foot long or a mile long. SFT had a complete, self-consistent blueprint of an electron with no scale bar on it.

Five separate searches confirmed this from five directions, and each closed a class rather than an instance:

- Stability arguments are homogeneous. Bundle transport, envelope balance, the closure spectrum — all fix shape, none fix size.
- Far-field gravity is blind to the interior. Two particles with equal mass and different internal calibrations produce identical exterior fields: a microscopic no-hair result.
- Near-field contact does not help. Resolving the two closure rings changes the force by five orders of magnitude, but every allowed gap rescales with the same ruler, so the ruler cancels exactly.
- Formation is scale-neutral. Conserved flux is unchanged under transverse rescaling, worldtube overlap fixes only a ratio, and the formation action is independent of the footprint radius outright.
- The write profile carries no information about its own normalization. Its shape is identical for every value of the constant it was supposed to determine.

Those are not five failures. They are one result stated five ways: the theory has no internal ruler, and its equations say so consistently, from every direction anyone has approached them. Which is what a relational account of gravity ought to say.

4 The register

What broke the deadlock was not another geometric argument. It was a counting argument.

A particle's support has a boundary, and that boundary is divided into discrete slots, each one primitive write long. How many slots go around it is not a matter of geometry. It is the capacity of the operator that maintains the particle — and that operator acts on a six-sector family register whose amplitude-transfer determinant is exact and independent of orientation:

$$D_6 = e^{3L_{\mathrm{eff}}} = 7.134970798893849 \times 10^{11}$$

One unit of charge walks that boundary exactly once, which fixes the normalization, and squaring the circumference-normalized capacity gives a baseline coefficient:

$$B_6 = \left(\frac{D_6}{2\pi}\right)^2 = 1.289509848425358 \times 10^{22}$$

Nothing in that expression is fitted. The dynamic scale L_{eff} was frozen versions earlier; the six sectors are the family structure; the 2π is one charge cycle. And the number lands within about three percent of what the electron requires.

Three percent, out of a determinant. That is the moment the problem stopped being "find a ruler" and became "explain a residual" — a far better problem, because residuals have causes.

5 What the electron withholds

The residual has a physical source, and it is the part of this result worth defending hardest.

The six-sector determinant belongs to the universal law — the rule that makes charged particles possible at all. It describes the whole register. But a particular electron does not occupy the whole register. It occupies one maintenance channel: the $n = 3$ weak lane, sitting exactly at the boundary between light and matter.

And that channel retains something. Each maintenance cycle, a fraction of the previous cycle's amplitude is carried forward rather than released. That retention is what gives a particle memory, and it is why the electron is the longest-memory and lightest charged channel there is.

The value of that fraction is where the result would be most vulnerable if it were an input — so it is worth showing that it is not one. The rigidity operator carries a lane term, $K = \frac{3\pi}{4} + \frac{L_{\text{eff}}}{4}u$, and the retained amplitude obeys $-\ln q = K\,P_{\text{coh}}$. At the exact matter/light boundary the threshold condition is $P_{\text{coh}}\,u = \frac{1}{3}$ — which makes the lane term exactly $L_{\text{eff}}/12$ no matter what u is. The lane variable cancels, and what remains is an equation that solves itself:

$$-\ln q = \frac{3\pi}{4}\,P_{\text{coh}}(q) + \frac{L_{\text{eff}}}{12}, \qquad q_* = 0.091710909655065$$

Three inputs, all frozen days earlier and none of them a long decimal: $3\pi/4$, $L_{\text{eff}}/12$, and the threshold $1/3$. The fifteen digits are not inherited from anywhere; they are manufactured by the root solve. The lane orientation is reconstructed *afterwards*, which is why correcting its published value at the fifteenth decimal place moved the final answer by one part in 10^{13}.

Retained amplitude is not available. It is already committed to the particle's own history, and it cannot be lent to the new boundary. Settlement has three directions and the electron occupies one, so the share withheld across the register is one third of that fraction — about 3.06 percent. Subtract it:

$$\chi_e = 1 - \frac{q_*}{3} = 0.969429696781645$$

That is the entire correction. Not a profile factor, not a fitted conversion, not a forgotten shape integral — those were each tried and each formally closed. It is the electron declining to lend out what it has already spent on being itself.

Universal register capacity, minus occupied retained capacity, equals the electron's capacity. The scalar that performs the subtraction is not chosen for its result: basis invariance, linearity across the settlement directions, continuity, and correct normalization select the normalized trace uniquely, and the alternatives — geometric mean, quadratic mean, determinant, the intensity-level version — miss by two hundred to sixty thousand times as much.

6 The number

The chain runs without further input:

$$\Gamma_e = \chi_e B_6 = 1.250089141355940{\times}10^{22} \quad \longrightarrow \quad \rho_e = 1.205621361455173{\times}10^{11} \quad \longrightarrow \quad \epsilon_e = 4.185464261523565{\times}$$

Then exactly one measured quantity enters, and it is not a gravitational one. The electron's inertial mass is determined in a Penning trap by electromagnetic means, to three parts in ten billion, with no gravitational measurement anywhere in its provenance. It converts the dimensionless share into SI units:

$$G_0 = \epsilon_e^2 \frac{\hbar c}{m_e^2} = 6.674306513706680 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$$

The 2022 CODATA recommended value is $6.674\,30(15) \times 10^{-11}$. The prediction sits 0.976 parts per million above the central value.

The context matters more than the agreement, and it cuts two ways.

First, G is the worst-measured constant in physics. Its recommended uncertainty is twenty-two parts per million, and the underlying experiments disagree with one another by roughly five hundred and fifty — badly enough that CODATA inflates every experiment's stated uncertainty by a factor of 3.9 before averaging them. So the prediction is about twenty times sharper than the number it is compared against, and five hundred times sharper than the spread of measurements underneath it.

Second, and less comfortably: the target has moved.

CODATA	central G (SI)	relative u (ppm)	G_0 − central (ppm)
2010	6.67384×10^{-11}	119.87	+69.90
2014	6.67408×10^{-11}	46.45	+33.94
2018	6.67430×10^{-11}	22.47	+0.976
2022	6.67430×10^{-11}	22.47	+0.976

The recommended value walked 68.9 parts per million between 2010 and 2018 — seventy times the residual now being reported — and it walked toward this prediction. Read optimistically, the measurements are converging on a value this formula would have named in 2010, when nobody could have fitted to it. Read sceptically, a target that soft should not be trusted to a part per million yet. Both readings are honest, and the same fact makes the claim more falsifiable rather than less: the prediction picks a specific point inside a disputed range, and sides with CODATA against the BIPM torsion-balance determinations some two hundred parts per million higher. **That is a bet, and it is the kind that gets settled by better metrology rather than by better averaging.**

7 What it means

Three things, in the order they should be defended.

Gravity's weakness is not a mystery about force strength. It is the ratio between one primitive write and the enormous span of history a particle lays down. Two electrons at their closest permitted approach defer roughly two writes in every 10^{45}. That is not a feeble force; it is an accurate accounting of how small a single write is compared to the record it belongs to.

The number called G is a unit conversion. The theory's actual output is the electron's position relative to the only mass scale at which gravity becomes visible on its own. Expressing that position in metres and kilograms produces the familiar numeral, and nothing more mysterious than that is happening. This is why no amount of internal cleverness was ever going to produce G directly, and why five independent no-gos found the same wall.

What a particle is made of and what it weighs are the same fact, counted twice. The register says how much boundary the law can support. The retained amplitude says how much of it this channel has already spent on its own past. The difference is the particle's size, and size is mass, because the carrier moves at one speed and cannot be sped up. There is no separate mass mechanism left to find.

8 What it does not mean

One postulate. The map from register capacity to physical coefficient is taken to be the identity — one unit of capacity buys exactly one unit of coefficient. Frozen doctrine permits any positive universal constant in that slot, and nothing currently forces unity. It is labelled a postulate in the source document and it should stay labelled. Worth knowing: no experiment can presently distinguish it. Because G depends on the inverse square of that constant, the twenty-two parts per million uncertainty in measured G constrains it only to about eleven, while the departure from unity that would give exact agreement is half a part per million — some twenty times inside the noise. Settling it would require G measured to about one part per million, a twentyfold improvement on anything achieved in two centuries of trying.

One species. This is a ground-state formula and it produces one particle. The retained fraction lies strictly between zero and one, so the correction can shave at most a third off the baseline, and the entire reachable mass window runs from about 0.97 to 1.45 electron masses. The electron and positron are inside it — and their exact mass equality comes out for free, since charge conjugation reverses the winding without changing the boundary. Nothing else in physics is inside it. The muon is 207 electron masses and the tau is 3477. This formula cannot reach them, and it was never going to.

Not blind, and not called blind. The programme was never ignorant of the measured value of G; no working physicist could be. What the receipt chain establishes is narrower and checkable: the candidate formula was generated from frozen internal equations before any numerical comparison. The source documents accordingly retired the phrase "blind derivation" in favour of "comparison-separated." The weaker word is the true one.

Not a confirmation. Agreement at one part per million with a quantity known to twenty-two is consistency, not verification. The prediction is currently sharper than the experiment that would test it — an unusual and rather good place for a theory to stand, but not the same as being confirmed.

9 The next question

The generation ladder does exist in the theory. The three lanes are derived, their rigidities stand in the exact ratio 1:4:1, and the operator that carries them is frozen. But every part of that structure currently reaches the mass through the same retained fraction — and that fraction is bounded. The theory has generations and the mass channel throttles them.

So the next question is not "find a generation mechanism." It is narrower and better: does the lane operator appear anywhere in the boundary construction, or only in the maintenance recurrence? If only the latter, that is the gap, and it is a specific one.

There is a further reason to want the answer. If a second route exists, the general coefficient becomes the present one multiplied by a lane-dependent factor — and the frozen electron result then demands that factor be exactly unity at the electron's lane, to within half a part per million, while suppressing the muon's by a factor of two hundred and the tau's by three thousand. That is a severe constraint. It also means the two programs are not independent: the particle atlas can falsify the gravity result.

A prediction that another arm of your own work could kill is a better object than one that sits safe. That is the position this result is in, and it is the right one to be in.

Status. Conditional numerical-G prediction frozen. Full derivation open by one universal optical normalization. Particle-geometry atlas active.

Originator and lead author: Shaun Fosmark. Derivation and audit canon: GPT-5.6 SOL. Independent branch critique and narrative: Opus. All numerical values are taken from the frozen V13.1-V3 Numerical G Companion v1.0 and its receipt chain.